

Safe Paths to School: Evidence from Bogotá’s Walking to School Caravans*

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Abstract

In many low- and middle-income countries, most children walk to school through dense and often unsafe neighborhoods. These commutes impose daily coordination burdens and safety risks on families. Yet despite its consequences for attendance, learning, and well-being, school commuting remains an overlooked dimension of education policy. This paper evaluates walking-to-school caravans (WSC), in which trained monitors accompany groups of children along safe routes to and from school. We study Bogotá’s Ciempiés program, launched in 2019 and rolled out across public primary schools. Using administrative data from 2016–2024 and a differences-in-differences design that exploits the staggered adoption of the program, we estimate causal effects on educational outcomes and road safety. Overall, WSC reduce school switching by 6.8 percentage points and branch switching by 10.9 points—over 40% declines from pre-treatment rates in the control group. We also find suggestive reductions in grade repetition and dropout, though no effects on traffic accidents. At USD 285 per child annually (between 24% and 51% cheaper than busing), WSC are a cost-effective, scalable intervention that eases family coordination burdens and improves school permanence and progression, making them a promising model for other dense urban settings worldwide.

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1 Introduction

School attendance is a prerequisite for learning, yet many children in Low- and Middle-Income Countries (LMICs) continue to miss school regularly due to logistical and coordination barriers in school transportation. In dense urban settings, where most children walk to school, commutes themselves can become a daily hurdle. Long distances, dangerous or high-traffic neighborhoods, and the absence of formal transport systems force families into difficult trade-offs: parents can accompany their children to school but lose hours of work, send them alone but worry about safety, or keep them home when coordination fails. Irregular attendance undermines learning outcomes (Aucejo and Romano, 2016; Liu et al., 2021), raising dropout risk (Koppensteiner and Menezes, 2021), and affecting peers (Goodman, 2014; Bennett and Bergman, 2021). Despite its importance, commuting has received far less attention in education policy than classroom inputs or curricula.

This paper evaluates a large-scale intervention designed to directly reduce commuting and coordination costs: Walking to School Caravans (WSC). These programs recruit trained adults to accompany groups of children along predetermined safe routes, with designated meeting points where parents can drop off and pick up their children. Children walk together, the youngest hand by hand, learning about traffic norms and safety along the way. WSC thus lower the cost of attendance by reducing the travel burden for parents, alleviating safety concerns for parents and children¹, and providing children with additional exercise and road safety knowledge.

We study one of the largest implementations of WSC, Bogotá’s *Ciempiés* (spanish for Centipede) program. Launched in 2019 and rolled out staggeredly across public primary schools, *Ciempiés* provides quasi-experimental variation in treatment timing. We combine rich administrative data from Bogotá’s Secretaries of Education and Mobility with a difference-in-differences design to measure the program’s impacts. We find that WSC can substantially improve children’s educational trajectories. Overall, enrollment in *Ciempiés* reduces school switching by 6.8 percentage points and branch switching by 10.9 points. We also find a reduction of 5 percentage points in grade repetition, though this estimate is imprecisely estimated. These effects are large in magnitude and suggest that WSC strengthen children’s attachment to their school. At a cost of only USD 285 per child per year, the

¹In Colombia, the setting of this study, about 18% of children report feeling unsafe on the way to school. Other Latin American countries perform equally poorly on this indicator, while the OECD average is around 8%. Safety concerns can lead to school absences and generate levels of stress that hinder learning. In fact, as Figure 1 shows, there is a negative correlation between the share of children who report feeling unsafe on the way to school and the average math score in PISA.

program is highly cost-effective compared to traditional mentoring or school transportation interventions.

Despite these educational benefits for children, we find no evidence that [WSC](#) improved road safety. Using traffic accident data, we compare treated and not-yet-treated census tracts and find no impact on accidents involving children under 12. School-issued accident alerts show similar null effects, if anything pointing to small increases in incidents. These results likely reflect the counteracting forces of safer routes and traffic education, on the one hand, and reduced adult supervision (two monitors for up to 50 children instead of parents), on the other. The value of [WSC](#) thus lies less in reducing accidents than in easing the daily coordination burden of getting children to school. In contexts where small disruptions in parents' schedules or transportation systems translate into lost school days, interventions like [WSC](#) can stabilize attendance and keep children in school.

The case of Bogotá illustrates why commuting is such a central barrier to education. In 2024, two-thirds of all school trips in the city were made on foot, while only 16 percent were made by bus and 9 percent by public transport. For children in public schools, who come from poorer families and are less able to pay for private transportation arrangements, walking is more predominant. Among families who do purchase transportation, the financial burden is substantial: households report average monthly expenditures of COP 170,000 (roughly USD 43), 1.5 times the cost of *Ciempies* on a per-child basis. This cost is particularly high when compared to Colombia's minimum wage, about USD 240 per month in 2023, meaning that formal transportation absorbs a large share of family income. For most low-income families, therefore, the only option is to accompany children by foot, rearranging work schedules and increasing the fragility of attendance.

Time burdens are also unevenly distributed across socioeconomic status. Data from Colombia's Quality of Life Survey ([QLS](#)) show that children from the lowest socioeconomic strata spend nearly 13 minutes walking to school on average, compared to under 10 minutes for children from higher strata. The difference may appear small, but multiplied across two daily trips and over the course of an academic year, it represents dozens of additional hours of commuting time borne disproportionately by the poorest families. In Bogotá, there's also a clear socioeconomic in school distance/travel time: children in extreme poverty (SISBEN score of 0) travel on average 2.2 kilometers to school, compared to 1.4 kilometers for children in less vulnerable households. Families in informal settlements often live farther from public schools and are more likely to be assigned to distant schools with available seats. These constraints mean that the children most at risk of dropout also face the most burdensome commutes.

Within this context, Ciempiés provides a community-based solution that reduces coordination costs and eases the burden of commuting with children for parents. By organizing children into supervised walking groups with designated meeting points, the program allows parents to delegate responsibility for part or all of the commute. This is particularly valuable given that the average route in the program covers 1.5 kilometers and requires about 40 minutes of walking time. For a working parent, this time commitment can make the difference between maintaining employment and having to forgo income. The program thus addresses not only safety concerns but also the opportunity costs of accompanying children, which are particularly high for parents in informal labor markets.

Traditional education policies have sought to increase attendance primarily through demand-side incentives such as Conditional Cash Transfers (CCT), which require 80–85 percent attendance and provide income support to poor families (Baird et al., 2014; Fiszbein et al., 2009; Millán et al., 2019). Other interventions reduce parental monitoring costs by providing information about attendance or assignments (Bergman, 2021; Berlinski et al., 2025; Lichand et al., 2024). Supply-side policies, such as school construction, reduce distances but are costly and slow to scale in urban areas (e.g., Burde and Linden, 2012; Dinerstein et al., 2023; Duflo, 2001; Khanna, 2022). Transport-based solutions represent a third approach, but their effectiveness is mixed. In high-income countries, busing has been widely used to increase racial integration, but recent evidence suggests it is not cost-effective: Angrist et al. (2022) find that busing in Boston and New York improves integration but has no effect on achievement, despite very high costs. In LMICs, bicycles have been shown to improve girls’ enrollment, test scores, and empowerment (Muralidharan and Prakash, 2017; Fiala et al., 2025), but they are limited to children old enough to ride and to areas with safe roads. WSC, by contrast, are low-cost, fast to scale, and well-suited to the realities of dense cities where most children walk to school.

The mechanisms through which WSC operate are multiple. They ease the time and coordination costs for families, particularly those with inflexible work schedules or multiple children to care for. They alleviate safety concerns by ensuring children are accompanied by trained monitors and peers. They provide children with daily exercise and practical knowledge of traffic rules. And they create a social commuting experience that may strengthen peer ties and children’s sense of belonging in school. These features distinguish WSC from other transport interventions, where the child is a passive rider rather than an active participant. We provide complementary survey evidence on each of these channels in Section 4.4, drawing on a 2025 end-of-program survey of program beneficiaries and their caregivers.

Our findings contribute to several strands of literature. We place commuting at the center of education debates, showing how overlooked coordination costs affect school attendance and permanence. We provide the first rigorous evaluation of WSC, complementing studies on bicycles in LMICs and buses in high-income countries (Angrist et al., 2022; Setren, 2024; Muralidharan and Prakash, 2017; Fiala et al., 2025). We add to research on urban safety and education, showing that programs designed to improve commuter safety for children may have limited effects when they substitute parental supervision (Bauernschuster and Rekers, 2022; Koppensteiner and Menezes, 2021). Finally, we contribute to the literature on neighborhood effects and place-based policies (Chetty et al., 2016; Chetty and Hendren, 2018a,b) by showing that intervening neighborhoods and creating safe paths to school can increase school attachment without requiring families to relocate away from their communities and social networks.

More broadly, our results highlight commuting as a critical but neglected margin of education policy. In cities where most children walk through dense and often dangerous neighborhoods, WSC represent a scalable model to improve student attendance and permanence at very low cost. Beyond easing parental coordination, they provide children with exercise and safety training and foster a stronger attachment to school. The lessons extend beyond LMICs: many developed countries face similar challenges in urban commuting, making WSC relevant as a complementary strategy to buses or other forms of school transport.

The rest of the paper proceeds as follows: Section 2 describes the setting and data. Section 3 outlines the differences-in-differences strategy, and Section 4 presents results. We conclude in Section 5.

2 Setting and Data

Our study focuses on Bogotá, the capital and largest city in Colombia, whose public school system served an estimated 1,142,000 preschool, primary, and secondary students in 2024. Families in Bogotá can choose between public and private schools, with public school admissions coordinated through a centralized assignment system. In 2024, 63% of students attended public schools, while 37% were enrolled in private institutions. The public school choice system transitioned from a Boston-style mechanism to a Deferred Acceptance algorithm during 2022–2023. Our analysis focuses exclusively on the public sector, as the *Ciempiés* program operates only in public schools.

Parents in Bogotá have numerous public school options, and while distance plays a role in school choice, families do not always choose the nearest public school. This behavior

reflects the consideration of multiple factors—such as school quality, perceived safety, and convenience regarding caregivers’ work.² Yet, this choice flexibility is not the same across socioeconomic groups, creating systematic disparities in school commuting patterns.

Figure 3 presents a binscatter regression that reveals a socioeconomic gradient in school distance. Children with lower SISBEN III scores, which indicate higher vulnerability, tend to live farther from their school. Those in extreme poverty (score = 0) travel, on average, 2.2 kilometers to school, while children with higher (though still vulnerable) scores travel around 1.4 kilometers. This pattern suggests that the poorest families face additional barriers to access, potentially due to informal housing in areas with poor public provision of education or late assignment to schools with available slots but in distant locations.³

The gradient in school distance translates into concrete burdens for vulnerable families. Figure 6 demonstrates that average travel times to school vary systematically by socioeconomic strata among children who commute by foot or bike. Children from the lowest strata (1 and 2) spend significantly more time commuting than those from higher strata, with strata 1 children requiring nearly 13 minutes on average compared to under 10 minutes for strata 4 children. These differences impose higher time costs and additional coordination challenges on low-income households, making consistent school attendance more challenging to maintain.

The transportation choices available to families reflect both preferences and constraints. Figure 4 illustrates that walking remains the most common way to commute to school in Bogotá, accounting for 66.4% of all school trips. This pattern is more pronounced in public schools, where students are significantly less likely to use paid transportation than their counterparts in private schools. The data show that only 16.4% of students use school buses, 9.4% rely on public transport, 4.7% travel by private vehicle, and 3.0% use bicycles.

For families who do pay for school transportation, the financial burden is substantial. Figure 5 presents the distribution of monthly transportation expenditures among households that report paying for school transportation services. On average, the amount that families pay per month for transportation services is approximately COP 170,000 (roughly USD 43). This amount is nearly double the cost that the city pays per child per month in *Ciempién* (USD 23.75), highlighting the program’s cost-effectiveness relative to private transportation alternatives.

²Many families, especially those with young children, enroll their children in schools close to their work-places to facilitate drop-off and pick-up.

³While in many LMICs richer families live farther from schools due to suburban housing patterns (Beuermann et al., 2023; Hofflinger et al., 2020), the inverse gradient in Bogotá reflects the concentration of urban poverty in poorly connected peripheries and the challenges of navigating the centralized assignment system.

The pattern that emerges is that many public school students live relatively far from school but cannot afford paid transportation options. This creates a challenge for low-income families, who must either invest significant time accompanying children on long walks to school or find alternative arrangements that may compromise safety or reliability.

In sum, while Bogotá offers choice within its public education system, geographic, logistical, and financial frictions constrain that choice for the poorest families. The consequences of this are longer travel times, greater need for coordination, and limited use of formal transportation services. This can lead to inconsistent school attendance and exacerbate inequality in access to education. The *Ciempiés* program operates within this context, attempting to address the coordination problem parents face in bringing their children to school every day by providing a community-based transportation solution.

2.1 The Ciempiés Program

Bogotá’s Secretaries of Education and Mobility launched *Ciempiés* in 2019. The program was designed to address safety concerns and coordination challenges faced by families whose children walk to school, particularly in unsafe areas and with high incidence of traffic.

The program operates through supervised walking caravans that transport children to and from school along predetermined safe routes. In each participating school, program staff map out an optimal walking path that avoids crime hotspots, minimizes exposure to heavy traffic, and connects the school with the highest concentration of children who walk to school. Two trained monitors lead organized walking caravans of up to 50 children on each route.

The key innovation of *Ciempiés* lies in its meeting point system, illustrated in Figure 2. Along each safe path, the program establishes designated meeting points where parents or caregivers can drop off and pick up children. This system reduces the coordination burden on families in several ways. Parents no longer need to accompany their children the entire way to school. Children benefit from adult supervision and peer interaction during the commute. Families gain flexibility in their daily schedules since the meeting points are strategically located to be accessible from multiple residential areas.

The program targets kindergarten and primary school children (ages 5-10) attending public schools. While there is an eligibility criterion requiring students to live within 700 meters of a meeting point, this criterion is not strictly enforced in practice, allowing program staff to accommodate families based on local needs and logistical considerations.

Ciempiés expanded through a staggered rollout that provides the quasi-experimental variation for this study. Starting with a pilot in 2018, and a first phase of the rollout in 2019,

the program has grown steadily, reaching 53 school shifts across 9 of Bogotá’s 19 localities by 2024. Figure 7 shows the spatial distribution of the program as of 2022. The program is scheduled to expand to an additional 75 schools through 2025-2030.

The cost of the program is approximately USD 285 per child per year (at 2023 prices), which includes monitor salaries, program administration, and equipment. This cost is significantly lower than formal transportation alternatives or intensive educational interventions, making it an attractive option for resource-constrained educational systems.

The relevance of this cost figure depends on the counterfactual against which the program is evaluated. End-of-program survey evidence collected from 945 caregivers of *Ciempiés* beneficiaries at the end of 2025 indicates that the program substitutes overwhelmingly for unpaid parental time rather than for paid transportation services: when asked how their child travelled to school *before* entering *Ciempiés*, 82.4% of caregivers report that the child walked, in nearly all cases accompanied by a parent or relative. Public buses, private vehicles, and paid school transport account for less than 11% of the distribution combined (Table 7). The appropriate benchmark for the program’s cost is therefore the shadow value of caregivers’ time rather than the COP 170,000 monthly out-of-pocket transportation expenditure documented above, and the program’s first-order effect operates through the coordination margin emphasized in the introduction.

2.2 Data

We combine administrative data from three institutions (two at the local and one at the national level). Our three data sources are the following:

Secretary of Mobility. From Bogotá’s Secretary of Mobility, we obtain detailed information about the Walking to School Buses (WSB) operation, including application and enrollment records for all participating children with their geocoded home addresses. The dataset includes daily attendance records for the walking caravans and documents which parent or caregiver dropped off and picked up each child at the meeting points.

Additionally, the Secretary of Mobility provided the universe of traffic accident reports in Bogotá from 2015-2024. These records include precise geolocation data, accident severity, victim demographics (age, gender), mode of involvement (pedestrian, passenger, driver, cyclist, motorcyclist), and timestamps. This comprehensive dataset allow us to examine effects of the program on accidents involving children of primary school age, and spillover effects to other age groups and geographic areas.

Secretary of Education. From Bogotá’s Secretary of Education, we access the SIMAT (Integrated System of Student Enrollment), an annual registry of all students enrolled in Colombian public schools from 2016-2024.⁴ This dataset includes socioeconomic information such as ethnicity, disability status, country of origin, socioeconomic stratum⁵, and SISBEN scores. The data allows us to look at outcomes like grade progression, school switching, and dropout.

We also have access to records of alternative mobility programs and transportation subsidies from 2022 to 2024. We merge these records with SIMAT data to obtain a comprehensive picture of the transportation assistance available to households. The other mobility programs are *Al Colegio en Bici*, a program that organizes caravans of children biking to school with bicycles lent by the city government, and *Biciparceros*, which organizes groups of children (older than 14) to go to school with their own bikes, providing supervision and mechanical assistance along the way. Figure A.4 shows the shares of children in the different mobility programs (including *Ciempiés* and transportation subsidies), by deciles of distance to school.

Our learning outcomes come from the *Plataforma de Apoyo Escolar* (2015-2024), a system that the Secretary of Education provides to schools for generating school transcripts. From these transcripts we retrieve the end-of-year math and language grades, and in some cases, teacher-recorded absences and measures of parental involvement. Teachers assign grades at their discretion within each school, but they must use a national scale with four achievement categories: low, basic, medium, and high.

Finally, we also have access to novel data on school issued alerts from the *Sistema de Alertas*, which schools have to report by law to the Secretary of Education. The data covers behavioral alerts, and violence and accident reports from 2014-2024, and includes the time and description of the event, and the identity of the aggressor, disruptive kid or victim involved. We use the alerts issued for any accident that happened on the way to school to complement our analysis on the impact of *Ciempiés* on road safety outcomes.

Instituto Colombiano para la Evaluación de la Educación (Icfes). From Icfes, we obtain standardized test scores from the *Pruebas Saber* 3, 5, and 9 for 2017. This was the last year that standardized tests were applied broadly to primary school students before an interruption

⁴For private schools, we only have data from 2020 onwards, since before that year they were not in SIMAT.

⁵The socioeconomic stratum, known as *estrato* in Spanish, refers to Colombia’s socioeconomic stratification system, which categorizes households based on their affluence using neighborhood and dwelling characteristics, with one corresponding to the poorest, and six to the richest households.

that lasted until 2022. After that year, only a representative sample of schools is tested.⁶ These data allow us to assess balance between schools that adopted *Ciempies* at some point between 2019 and 2024 and future adopters (never-treated group) before the program began, providing an indirect way to evaluate parallel trends.

Table 1 summarizes the characteristics of children participating in *Ciempies* from 2021-2024. Participants are typically 8 to 9 years old (corresponding to third and fourth grades). Half of them are girls. The average child in our sample comes from socioeconomic stratum 2, indicating these are vulnerable children from low-income families. The data also reveal a higher representation of Venezuelan children and victims of Colombia’s armed conflict, particularly in recent years. These groups face additional barriers to accessing education. For comparison, 7% of Bogotá’s public school students were Venezuelan, and 4% were victims of Colombia’s armed conflict in 2024, versus 13% and 8% in our sample.

Table 2 provides descriptive statistics for the walking paths from 2020-2024. The average path length is approximately 1.5-1.6 kilometers, requiring 40-42 minutes of group walking time. The program serves different school shifts, with a relatively balanced distribution between morning and afternoon shifts, plus some full-day programs.

These path characteristics highlight several important features of the program. First, the walking distances are substantial, which underscores why parents may have safety concerns and used to accompany their children to school or arrange a transportation service in the absence of the program. Second, the 40+ minute (one way) walking time represents a considerable daily time investment for families, making the supervised caravan model particularly valuable for working parents who cannot accompany children for such extended periods.

3 Empirical Strategy

To estimate the causal impact of *Ciempies* on children’s educational outcomes and road safety, we exploit the program’s staggered rollout across Bogotá’s public schools between 2019 and 2024. We implement a Difference-in-Differences (DiD) approach using the methodology proposed by Callaway and Sant’Anna (2021), which allows us to estimate group-time Average Treatment Effect on the Treated (ATT) for cohorts of children that began participating in different years.

⁶The *Pruebas Saber* evaluate student competencies in math, language, natural sciences, and citizenship across 3rd, 5th, and 9th grades. The 2017 edition only administered language and math tests. Individual test scores are categorized into four performance levels: insufficient, minimum, satisfactory, and advanced. Table A.2 describes each of these categories.

This method compares the evolution of outcomes for treated children in schools where *Ciempiés* started operating relative to children in schools that have not yet implemented the program, but where the program expanded later or is set to expand in 2025-2030. Specifically, for each treatment cohort g and year t , we estimate:

$$ATT(g, t) = \mathbb{E}[Y_{it} - Y_{ig-1} \mid G_i = g] - \mathbb{E}[Y_{it} - Y_{ig-1} \mid G_i = g'], \text{ with } g' > t \quad (1)$$

where Y_{it} is the outcome of interest (e.g., dropout, grade repetition, switching schools or school branches, math and language grades) for student i in year t , and $G_i = g$ indicates the year cohort g in which the student started treatment. We use as controls all students in schools that have not been treated yet by time t (so cohorts $g' > t$). For the earlier cohorts, these include not-yet treated, and never-treated students, while for the last treated cohort (the one that started with *Ciempiés* in 2024), these include only never-treated students ($g = \infty$). In Equation(1), we include age and gender as controls.

The main assumption we have to make for the internal validity of our estimates is the parallel trends assumption based on not-yet-treated units, which states that in the absence of treatment, average potential outcomes of the group first treated in g and the group not-yet-treated in $t > g$, would have evolved in parallel.

We compute event-study estimates by grouping treatment effects across different cohorts by the number of years that have passed since program implementation:

$$ATT_t^w = \sum_g w_g \cdot ATT(g, g + l) \quad (2)$$

where l indexes the number of years since treatment began, and w_g represents the relative size of cohort g . This aggregation enables us to examine dynamic treatment effects and assess whether the program's impact grows or fades over time.

In our main specifications, we focus on a set of educational outcomes at the individual level using SIMAT: school switching (including switching between branches or *sedes*), grade repetition, and dropout. We measure dropout as a dummy variable that takes the value of one if, by the end of school year t , the student i didn't enroll in the public school system the following year. We define grade repetition in t as being in the same grade as in $t - 1$.

To estimate the unintended effects on traffic accidents, we apply a similar DiD framework to traffic accident data but at the census tract level. We compare the number of accidents in census tracts intersecting with operating school paths to those in census tracts that in-

tersect not-yet-treated school paths.⁷ Additionally, we expand the not-yet-treated control group to include a never-treated control group consisting of all untreated census tracts in the *localidades* where *Ciempiés* operates.

Since our data on accidents is anonymized, we can’t establish whether the accident victim is attending any of the schools in our sample. As such, our estimates recover both the effect of the program and any spillover effects that may occur if other pedestrians or drivers change their behavior along the safe paths to school. For this reason, we complement our analysis by focusing on the school issued alerts on student road accidents, which schools need to report to the Secretary of Education. We estimate Equation 1 at the school level for all the periods before and after program implementation, and for all cohorts, and also present these results in the next section.

4 Results

Our empirical strategy provides the causal impact of the *Ciempiés* WSB program on children’s educational outcomes and road safety. Before turning to our main results, we examine whether schools that adopted *Ciempiés* at some point between 2019-2024 differed systematically from the not-yet adopters where the program will expand in the future. Table 5 reports means and standardized differences of baseline covariates measured in 2017-2018 (before any school adopted the program), including demographics, socioeconomic strata, traffic accidents on the way to school, and standardized test scores from the 2017 Saber assessments. We present both unweighted and student-weighted normalized differences.

The balance in covariates between adopters and non-adopters provides an indirect way of assessing the plausibility of the parallel trends assumption (Baker et al., 2025). If treated and control units are similar in observable variables that explain potential education outcomes and trends, this strengthens the claim that potential outcomes would have evolved in parallel in the absence of treatment. Our results in Table 5 show relatively small standardized differences between adopters and non-adopters, with most falling below 0.25 in absolute value—a commonly used threshold for meaningful imbalance (Baker et al., 2025). This evidence supports the credibility of our identification strategy and complements the graphical inspection of pre-trends that we discuss below.

⁷We use a catchment area of 250m around school paths to define treated versus control census tracts. Figure 8 illustrates the procedure that we follow.

4.1 School Switching

Before presenting the causal estimates, we document that school and branch switching are associated with worse subsequent educational trajectories. Tables 3 and 4 report linear probability model estimates of future dropout on lagged switching indicators, controlling for school $\times$ grade fixed effects. A student who switched schools in the current or prior year is 2–3 percentage points more likely to drop out the following year; a student who switched branches is 2 percentage points more likely to drop out. These associations survive the inclusion of individual demographic and socioeconomic controls and are precisely estimated across more than three million student-year observations. We emphasize that these regressions are purely descriptive—switching may be capturing underlying family instability or dissatisfaction rather than causing dropout directly. Nevertheless, they establish that the switching outcomes we study are not innocuous: they are predictive of children leaving school, making reductions in switching a meaningful outcome in this setting.

We now turn to the main results from our DiD estimation using the Callaway and Sant’Anna (2021) methodology, exploiting the staggered rollout of the program across Bogotá’s public schools between 2021 and 2024.

Figure 9 presents our dynamic estimates of the impact of *Ciempies* on children’s educational trajectories. The corresponding estimates are reported in Table 6.

The event-study plots in Figure 9 show the dynamic ATT for four key outcomes: dropout, grade repetition, school switching, and switching between school branches. Each panel displays treatment effects relative to the year before program implementation, aggregated among the different cohorts, with negative time periods indicating pre-treatment periods and positive values representing post-treatment effects. Since the 2021 cohort is the only one we can observe for three years after treatment, and it is the smallest ($N = 771$), our results for $t = 3$ (and $t = 2$ for dropout) are noisy.

Panel C of Figure 9 shows that *Ciempies* significantly reduces school switching. Overall, *Ciempies* decreases school switching by 6.8 percentage points. Relative to the pre-treatment school switching rate of 16.5% in the control group, these represent a reduction of 41%. We find even more pronounced effects for the switching between school branches variable. Figure 9, Panel D shows that *Ciempies* reduces branch switching by 10.9 percentage points. Compared to the pre-treatment branch switching rate of 25.8% in the control group, these effects represent a reduction of 42%. These large effects suggest that the program substantially increases children’s attachment to their school by reducing transportation coordination

challenges, which often force families to change their children’s schools from one year to the next.⁸

The pre-treatment coefficients in both panels show no systematic differences between treatment and control groups before program implementation, supporting the parallel trends assumption underlying our identification strategy. The growing magnitude of effects over time suggests that the benefits of the program compound as families become more accustomed to the *Ciempies* routine and the social bonds within the walking caravans strengthen.

4.2 Grade Progression and Dropout

For grade repetition and dropout, shown in Figure 9, Panels A and B, respectively, we observe point estimates suggesting program benefits, though these effects are imprecisely estimated with large confidence intervals. Panel B shows that *Ciempies* reduces grade repetition by 5 percentage points. Compared to the pre-treatment grade repetition rate of 2.3% in the control group, this reduction is substantial, though the statistical imprecision warrants caution in interpretation. Panel A suggests the program reduced dropout by 1.3 percentage points in the year it was adopted ($t=0$), representing a 13% reduction from the pre-treatment dropout rate of 9.75% in the control group. Although this effect is statistically significant, the coefficients for $t=1$ and $t=2$ jump around zero, resulting in an overall effect of 1.4 percentage points that is not statistically different from zero. These results reflect our limited power for long-term outcomes.

4.3 Results on Accidents

Although road safety is a stated objective of the *Ciempies* program, we find limited evidence of effects on traffic accidents. We study accidents using administrative records at the census tract level, comparing tracts that intersect with active *Ciempies* school paths to tracts that intersect with school paths that have not yet been treated, using a not-yet-treated baseline following Callaway and Sant’Anna (2021).

Our main specification excludes the years 2020 and 2021. During this period, schools were largely closed, and the program did not operate as a walking-to-school caravan, so children were not regularly commuting to school. Including these years yields qualitatively similar but substantially noisier estimates. We therefore focus on the specification that aligns treatment timing with periods in which the program was actively operating.

⁸Appendix Figure A.5 shows that the reductions in school and branch switching are concentrated among boys and among children living above the median distance to school.

Figure A.3 shows the treatment rollout across cohorts under this definition, while Figure 11 illustrates the spatial and temporal distribution of treated census tracts.

Figure 13 presents event-study estimates for traffic accidents involving children aged 5 to 12, the age group most closely aligned with program participation. The event-study coefficients are centered around zero in both the pre- and post-treatment periods, and confidence intervals are wide. We find no statistically significant effects of *Ciempiés* on accident incidence. While imprecisely estimated, the point estimates a few years after the program starts are generally negative, suggesting that accidents may decline with sustained exposure. As in our main educational outcomes, however, these later-period estimates are based on fewer treated cohorts and should be interpreted with caution.

In contrast, Figure 12 presents event-study estimates for accidents involving adolescents aged 13 to 18. For this group, which is not directly targeted by the program, we find no systematic post-treatment pattern. Estimated coefficients remain close to zero throughout the event window, with no indication of delayed effects.

Taken together, these results suggest that *Ciempiés* does not generate large changes in traffic accidents. At the same time, the absence of any increase in accidents is informative. The program substantially alters the organization of school commutes: under *Ciempiés*, two trained monitors accompany large groups of children along predefined routes, whereas in the absence of the program children are often accompanied individually by parents or caregivers. Despite this change in supervision structure, we find no evidence of increased accident risk and some suggestive evidence of a decline in accidents among younger children a few years after the program starts.⁹

As in the rest of our analysis, these findings point to the program’s primary effects operating through improved coordination and reduced logistical burdens for families rather than through large, directly measurable improvements in safety outcomes. The accident results, therefore, complement the main findings of the paper by showing that the gains in school attachment documented in Section 4.1 are not accompanied by adverse effects on child safety during the school commute.

⁹Evidence from surveying 945 caregivers and 363 students currently enrolled in the program is consistent with this reading. On a one-to-ten scale, average caregiver-rated quality of the monitor service is 9.81 and average student-rated perceived safety with the monitors is 9.71, with at least 97% of responses in both groups at eight or higher; one in three caregivers spontaneously chose, when asked about fears regarding the walk to school, an option indicating that the program had demonstrated to them that there was no longer reason for concern. We discuss these survey results in detail in Section 4.4.

4.4 Mechanisms: Survey Evidence

To complement the administrative-data results, we use surveys collected by the Secretary of Education of Bogotá in November and December of 2025 from 945 caregivers and 363 children currently enrolled in *Ciempies*.¹⁰ The survey speaks to three claims made elsewhere in the paper.

Parent rotation at the meeting point. The meeting-points appear to enable mutual-aid arrangements among program-using families. About one in four caregivers reports being responsible at the meeting point not only for their own child but also for additional children of other program-using families. Among those caregivers, the median caregiver supervises two additional children—that is, three program children in total at the meeting point (Table A.1, Panel A.3).¹¹ Because the meeting points are fixed in time and place and are visited daily, they appear to lower the cost for caregivers to coordinate informal pickup and drop-off rotations: rather than each family sending an adult twice a day, neighbouring families pool that responsibility, and a single caregiver can cover for several program children at once. This is consistent with the coordination-cost mechanism emphasized in the introduction and complements the formal monitor accompaniment by reducing the parental time burden across households.

Subjective safety. Section 4.3 attributes the null effect on traffic accidents to offsetting forces—safer routes and traffic education on one side, fewer adult eyes per child on the other—rather than to any deterioration in safety. Survey responses support that interpretation. On a one-to-ten scale, the average caregiver rates the monitor service at 9.81, and the quality of caregiver–monitor interaction at 9.67. Among children, average perceived safety while accompanied by the monitors is 9.71. When caregivers were asked, in an open-ended question, what they fear about the school commute, one in three spontaneously selected an option indicating that the program had demonstrated to them that there was no longer reason for concern (Table A.1, Panel B.1). These are perception variables and cannot speak to the actual level of safety, but they document the absence of any subjective deterioration that could be a concern in light of the null result on accidents.

¹⁰The surveys were administered through the Secretary of Education’s online platform. Because the instrument was administered to currently enrolled beneficiaries, the results describe attachment among the retained and should not be read as a causal test of attachment itself; switchers and dropouts are mechanically absent from the sample.

¹¹The survey item is conditional: caregivers were first asked whether they were responsible at the meeting point for any students besides their own children, and only then asked how many. We therefore interpret the count as a count of *additional* children, on top of the caregiver’s own.

Perceived behavioural changes and student-reported reasons. The program’s main effect—a forty-one to forty-two percent reduction in school and branch switching—is interpreted in Section 4.1 as evidence of stronger attachment to school. The survey responses speak most directly to that interpretation. When caregivers are asked which behavioural changes they have observed in their child since enrolling in *Ciempies*, the modal responses are that the child *goes to school with enthusiasm* (17.4%), *is happier* (14.3%), *is more responsible* (12.3%) and *has learned traffic signals* (11.4%). Children themselves report a mean happiness with the walk of 9.58 out of 10, and 97.8% of children and 98.2% of caregivers want the program to continue the following year. In addition, as Figure 15 shows, when children are asked which features of the program they enjoy the most, the share selecting each option maps almost one-to-one onto the three student-facing mechanisms named in Section 1: safety; knowledge and exercise; and peers and belonging. Together, Table A.1 and Figure 15 provide some suggestive evidence on the mechanisms behind the school-switching results.

5 Conclusion

This paper shows that relatively simple, community-based interventions can meaningfully improve educational outcomes by addressing the practical coordination challenges that families face when accessing education. The *Ciempies* program in Bogotá successfully reduced school and branch switching by up to 6.8 and 10.9 percentage points, respectively, after two years in the program. These effects represent 41% and 42% reductions from pre-treatment rates in control schools, indicating that the program significantly increases children’s attachment to their schools and reduces the disruptions that often force families to seek alternative schooling options.

While we find suggestive evidence of reductions in grade repetition and dropout, these effects are imprecisely estimated, likely reflecting our limited sample size for long-term impacts. Importantly, we do not find evidence that the program improved road safety for children, suggesting that the primary value of *Ciempies* lies in addressing coordination and logistical barriers rather than direct safety improvements. At a cost of just USD 285 per child per year, the program is a highly cost-effective way to improve school retention compared to intensive, individualized interventions or formal transportation alternatives. Survey evidence further indicates that the program substitutes for unpaid parental accompaniment rather than for paid transportation services, so the appropriate benchmark for its cost is the shadow value of caregivers’ time, and the meeting points additionally appear to enable informal pickup-and-drop-off rotations among program-using families, further reducing the parental time burden at no marginal cost to the program. These findings highlight the potential of place-based poli-

cies that improve educational opportunities within existing communities without requiring families to relocate or children to change schools, offering a promising model for addressing educational inequality in other urban contexts across [LMICs](#).

References

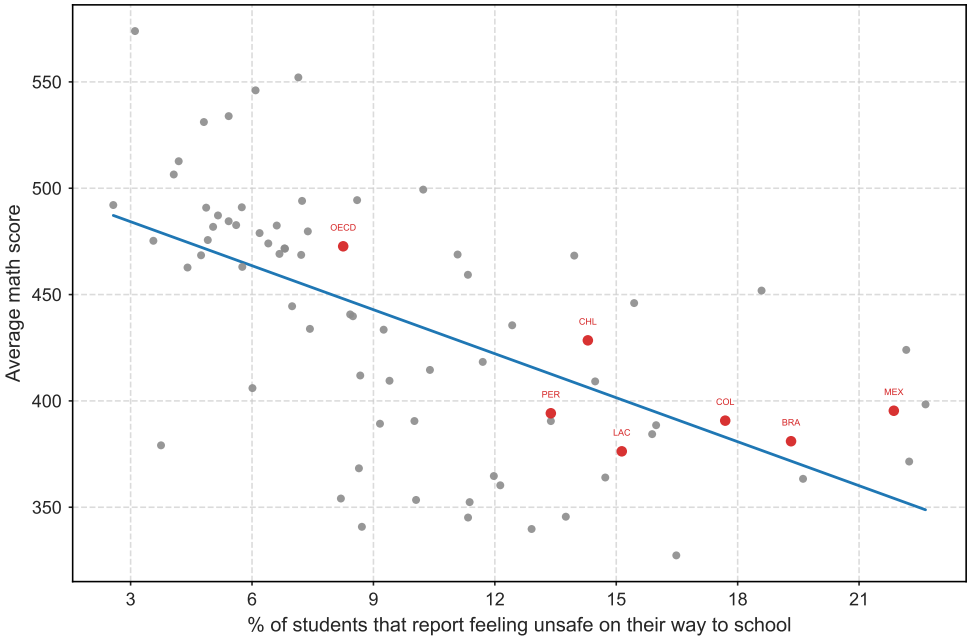
- Angrist, J., G. Gray-Lobe, C. M. Idoux, and P. A. Pathak (2022). Still Worth the Trip? School Busing Effects In Boston and New York. Technical report, National Bureau of Economic Research.
- Aucejo, E. M. and T. F. Romano (2016). Assessing the Effect of School Days and Absences on Test Score Performance. *Economics of Education Review* 55, 70–87.
- Baird, S., F. H. Ferreira, B. Özler, and M. Woolcock (2014). Conditional, unconditional and everything in between: a systematic review of the effects of cash transfer programmes on schooling outcomes. *Journal of Development Effectiveness* 6(1), 1–43.
- Baker, A., B. Callaway, S. Cunningham, A. Goodman-Bacon, and P. H. C. Sant’Anna (2025). Difference-in-Differences Designs: A Practitioner’s Guide. *Journal of Economic Literature*. Forthcoming.
- Bauernschuster, S. and R. Rekers (2022). Speed limit enforcement and road safety. *Journal of Public Economics* 210.
- Bennett, M. and P. Bergman (2021). Better Together? Social Networks in Truancy and the Targeting of Treatment. *Journal of Labor Economics* 39(1), 1–36.
- Bergman, P. (2021). Parent-Child Information Frictions and Human Capital Investment: Evidence From a Field Experiment. *Journal of Political Economy* 129(1), 286–322.
- Berlinski, S., M. Busso, T. Dinkelman, and C. Martínez A. (2025). Reducing Parent–School Information Gaps and Improving Education Outcomes: Evidence from High-Frequency Text Messages. *Journal of Human Resources* 60(4), 1284–1322.
- Beuermann, D. W., C. K. Jackson, L. Navarro-Sola, and F. Pardo (2023). What Is a Good School, and Can Parents Tell? Evidence on the Multidimensionality of School Output. *The Review of Economic Studies* 90, 65–101.
- Burde, D. and L. L. Linden (2012). The Effect of Village-Based Schools: Evidence From a Randomized Controlled Trial in Afghanistan.
- Callaway, B. and P. H. Sant’Anna (2021). Difference-In-Differences with Multiple Time Periods. *Journal of Econometrics* 225(2), 200–230.
- Chetty, R. and N. Hendren (2018a). The Impacts of Neighborhoods on Intergenerational Mobility I: Childhood Exposure Effects. *The Quarterly Journal of Economics* 133(3), 1107–1162.

- Chetty, R. and N. Hendren (2018b). The Impacts of Neighborhoods on Intergenerational Mobility II: County-Level Estimates. *The Quarterly Journal of Economics* 133(3), 1163–1228.
- Chetty, R., N. Hendren, and L. F. Katz (2016). The Effects of Exposure to Better Neighborhoods on Children: New Evidence from the Moving to Opportunity Experiment. *American Economic Review* 106(4), 855–902.
- Dinerstein, M., D. Morales, C. Neilson, and S. Otero (2023). Equilibrium effects of public provision in education markets. Working paper (Work in progress), September 30, 2023.
- Duflo, E. (2001). Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence From an Unusual Policy Experiment. *American Economic Review* 91(4), 795–813.
- Fiala, N., A. Garcia-Hernandez, K. Narula, and N. Prakash (2025, May). Wheels of Change: Transforming Girls’ Lives with Bicycles. *Economic Journal*. Forthcoming.
- Fiszbein, A., N. Schady, F. H. Ferreira, M. Grosh, N. Keleher, P. Olinto, and E. Skoufias (2009). *Conditional Cash Transfers: Reducing Present and Future Poverty*. World Bank Policy Research Report. Washington, DC: World Bank. License: CC BY 3.0 IGO.
- Goodman, J. (2014). Flaking Out: Student Absences And Snow Days as Disruptions of Instructional Time. NBER Working Paper No. 20221.
- Hofflinger, A., D. Gelber, and S. Tellez Cañas (2020). School Choice and Parents’ Preferences for School Attributes in Chile. Technical report.
- Khanna, G. (2022). Large-Scale Education Reform in General Equilibrium: Regression Discontinuity Evidence From India. Technical report. Publisher: The University of Chicago Press.
- Koppensteiner, M. F. and L. Menezes (2021). Violence and Human Capital Investments. *Journal of Labor Economics* 39(3), 787–823. Publisher: The University of Chicago Press.
- Lichand, G., N. Cunha, R. A. Madeira, and E. Bettinger (2024). When the Effects of Informational Interventions Are Driven by Salience: Evidence from School Parents in Brazil. Working paper.
- Liu, J., M. Lee, and S. Gershenson (2021). The Short- and Long-Run Impacts of Secondary School Absences. *Journal of Public Economics* 199, 104441.

- Millán, T. M., T. Barham, K. Macours, J. A. Maluccio, and M. Stampini (2019). Long-Term Impacts of Conditional Cash Transfers: Review of the Evidence. *The World Bank Research Observer* 34(1), 119–159.
- Muralidharan, K. and N. Prakash (2017). Cycling to School: Increasing Secondary School Enrollment for Girls in India. *American Economic Journal: Applied Economics* 9(3), 321–50.
- Setren, E. (2024). Busing to Opportunity? The Impacts of the METCO Voluntary School Desegregation Program on Urban Students of Color. NBER Working Paper No. 32864.

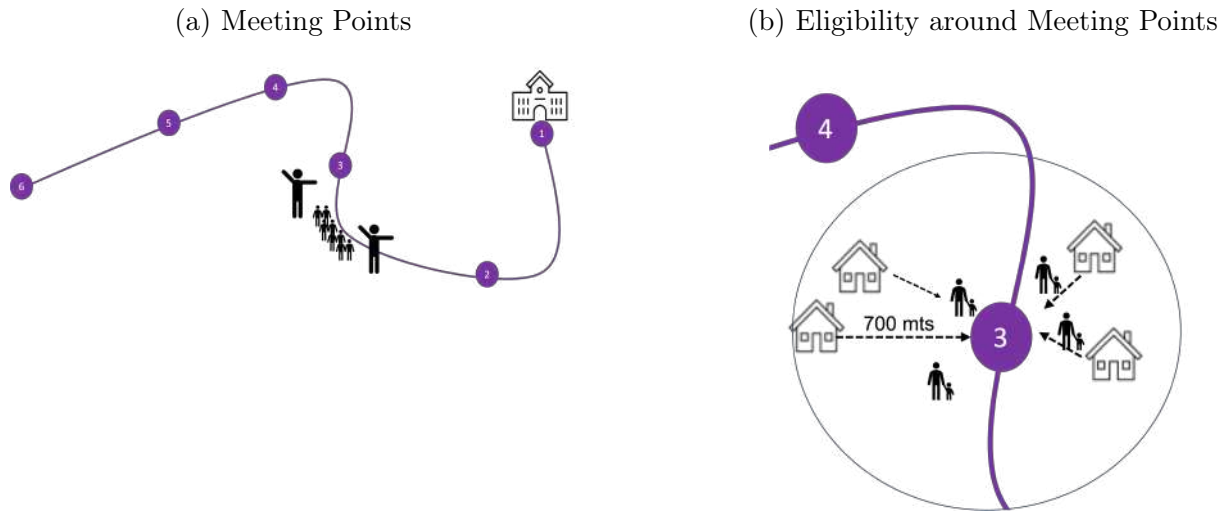
Figures and Tables

Figure 1: Perceived Unsafe School Commute and Average Math Scores



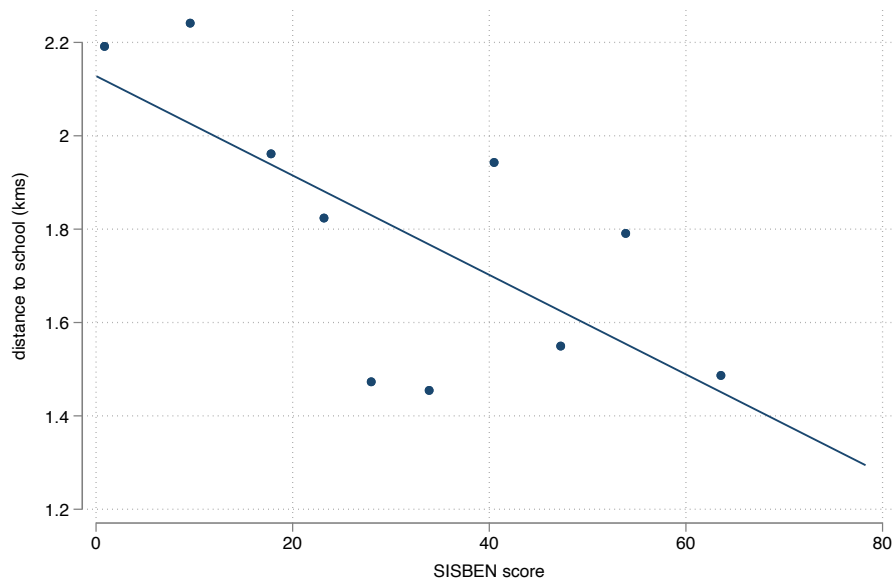
Notes: This scatterplot shows the relationship between average math scores and the share of students who report feeling unsafe on their way to school using data from PISA 2022. Countries from Latin America and the Caribbean (LAC) in the sample (Brazil, Chile, Colombia, Mexico, and Peru) are highlighted in red, along with the LAC and OECD averages. The linear regression fit is shown in blue.

Figure 2: WSC Operation



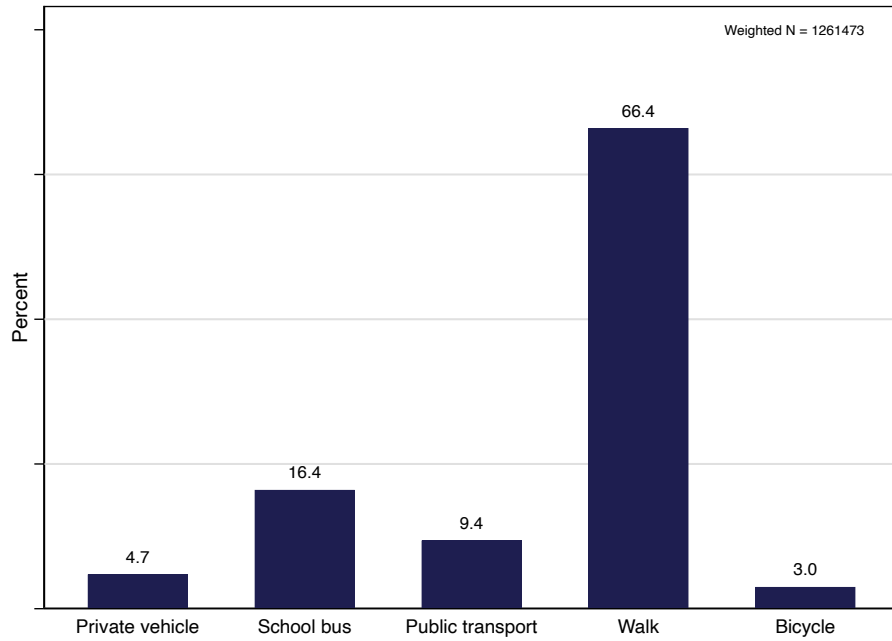
Notes: This figure shows a depiction of the WSC operation. Panel A shows the meeting points along the safe path to school. Panel B shows how a meeting point is used by the households living close by to drop and pick up children. Although there is an eligibility criterion on paper that says that students need to live within a 700m radius of any meeting point, this criterion is not enforced in practice.

Figure 3: Binscatter Regression of Distance to School on SISBEN Score



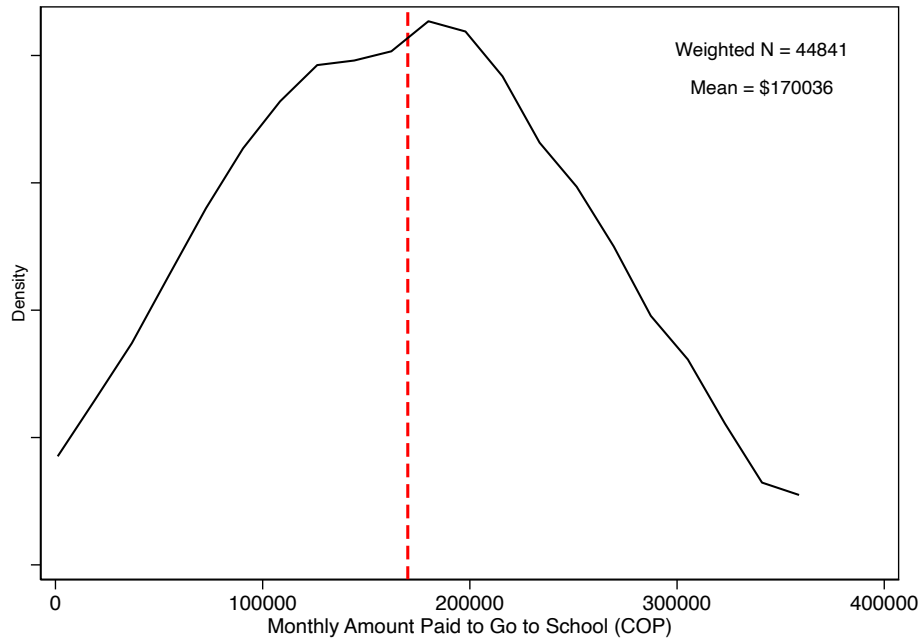
Notes: This figure shows the relationship between distance to school and SISBEN score, a household vulnerability index used to target social programs in Colombia. The sample corresponds to students in *Ciempies* schools in 2021. The SISBEN score ranges from 0 to 100, with lower scores indicating higher vulnerability, and a score of 0 indicating extreme poverty.

Figure 4: School Transportation Modes in Urban Bogotá



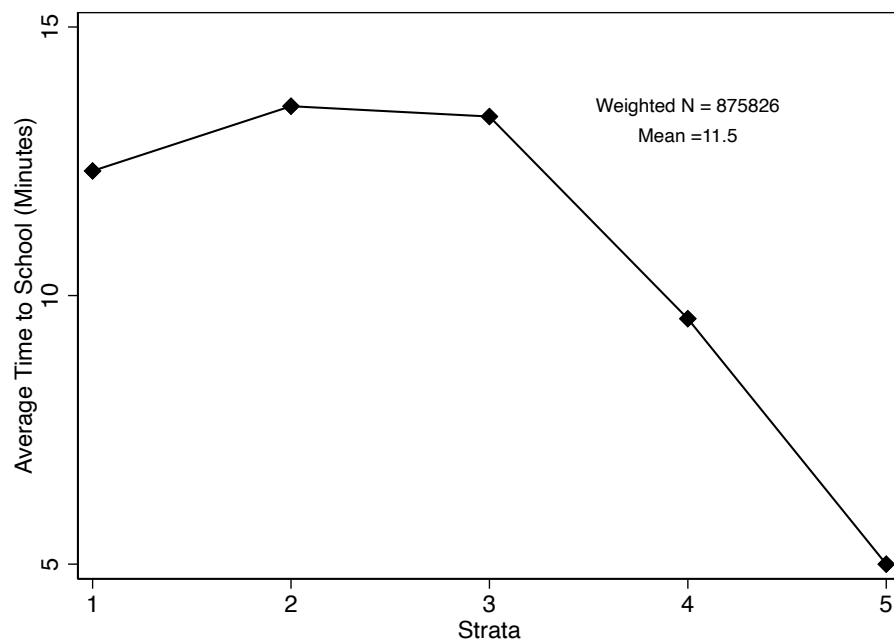
Notes: This figure shows the different transportation modes for children in public and private schools in Bogotá. The data corresponds to Colombia's 2024 [QLS](#).

Figure 5: Commuting Costs to Schools



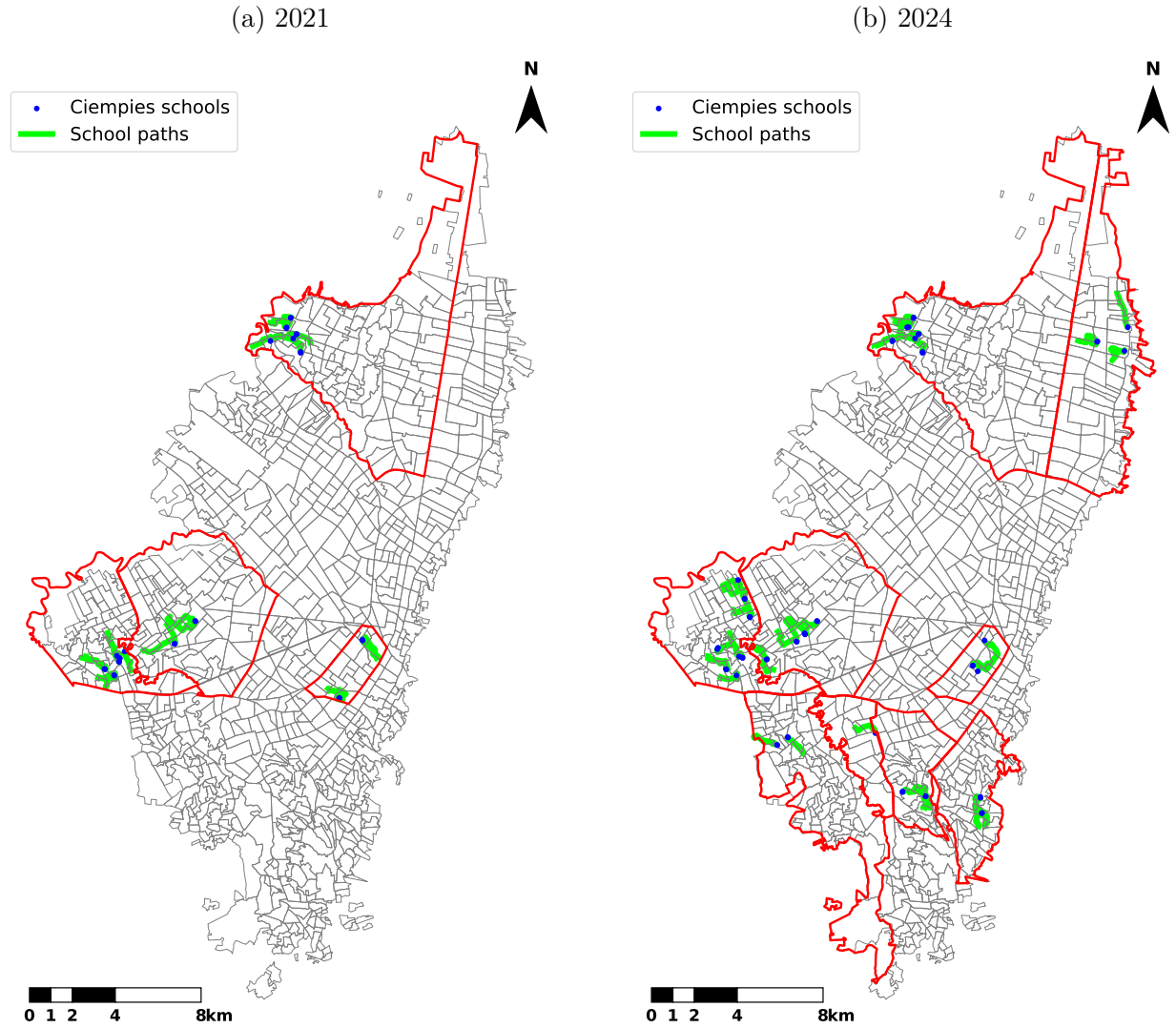
Notes: This figure shows the distribution of monthly commuting costs to school for the households that report paying for a transportation service. The dashed red line corresponds to the mean. The data corresponds to Colombia's 2024 [QLS](#). All values are in COP.

Figure 6: Commuting Time to School by Socioeconomic Stratum



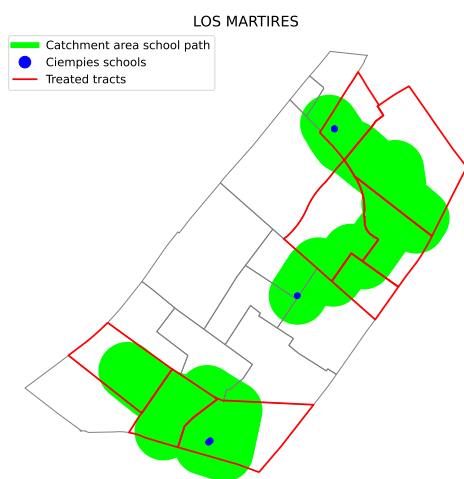
Notes: This figure shows the relationship between commuting time to school and socioeconomic stratum for children that commute to school by bike or on foot. The data corresponds to Colombia's 2024 [QLS](#).

Figure 7: Ciempiés Schools and Paths 2021-2024



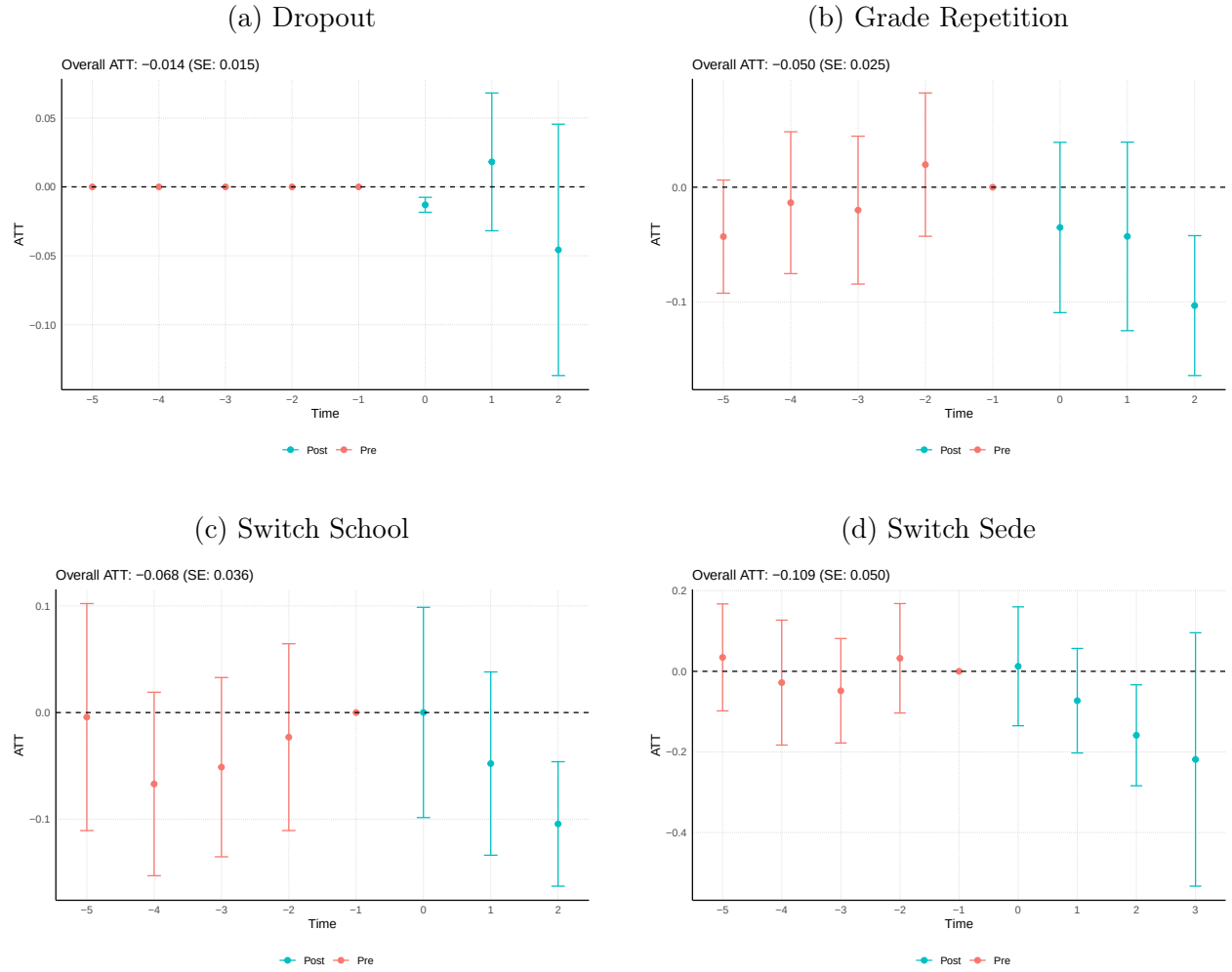
Notes: This figure shows the spatial distribution of *Ciempies* schools and paths through the program expansion between 2021 and 2024. In 2021, the program was operating in only 4 of the 19 *localidades* in Bogotá (pictured in red), but has since expanded its operations to 9 *localidades* as of 2024.

Figure 8: Defining Treatment and Control Census Tracts for Traffic Accident Analysis



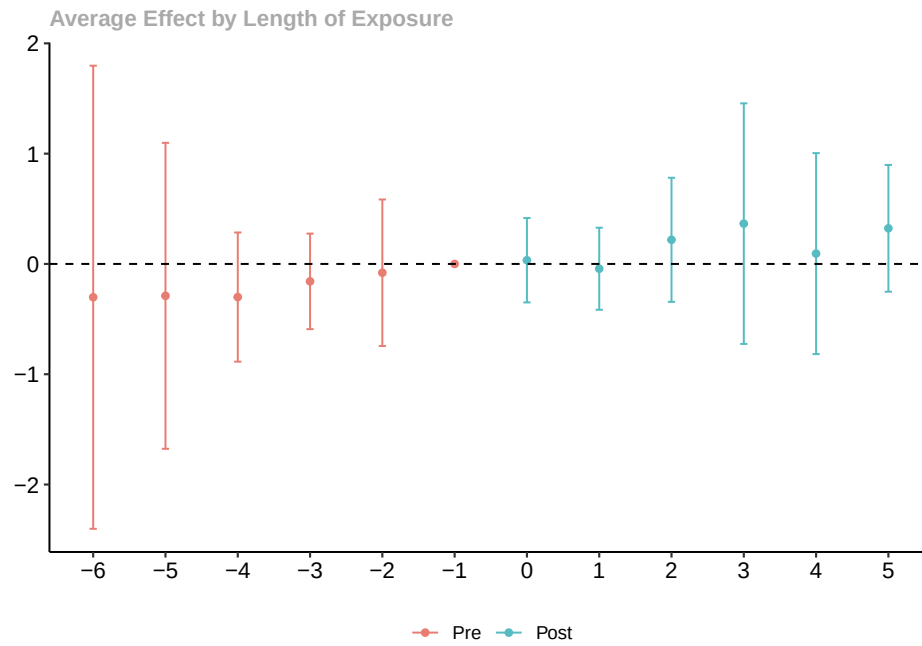
Notes: This figure shows the procedure to define treated versus control census tracts using catchment areas around school paths. We define a treated census tract if its area intersects the catchment area of a school path in at least 50%.

Figure 9: Impact of Ciempiés on School Trajectories of Children



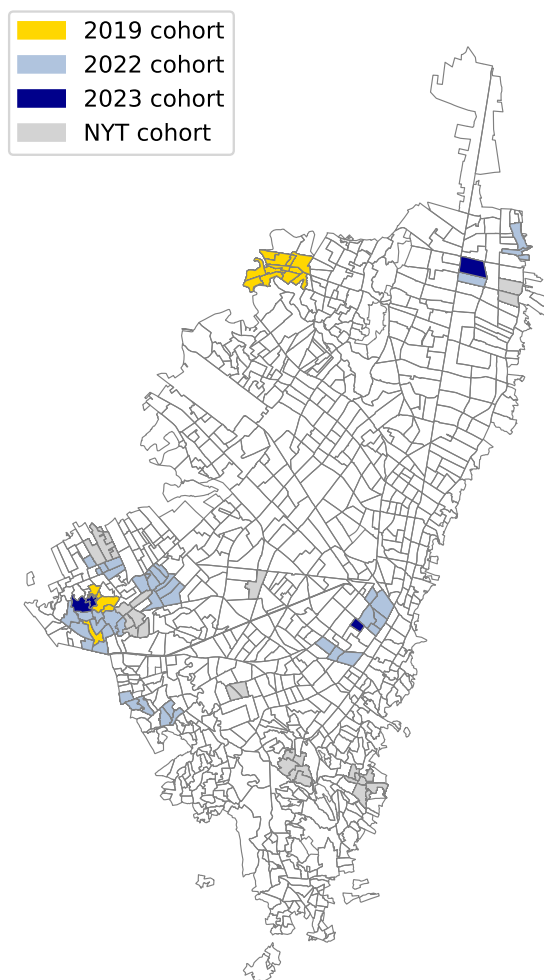
Notes: This figure shows the event-study aggregation of the estimated ATT for dropout, grade repetition, school switching and sede switching. The control group are the not yet treated students and the never treated students in school where the program is set to expand. 95% confidence intervals reported.

Figure 10: Results for Road Accidents at the School Level



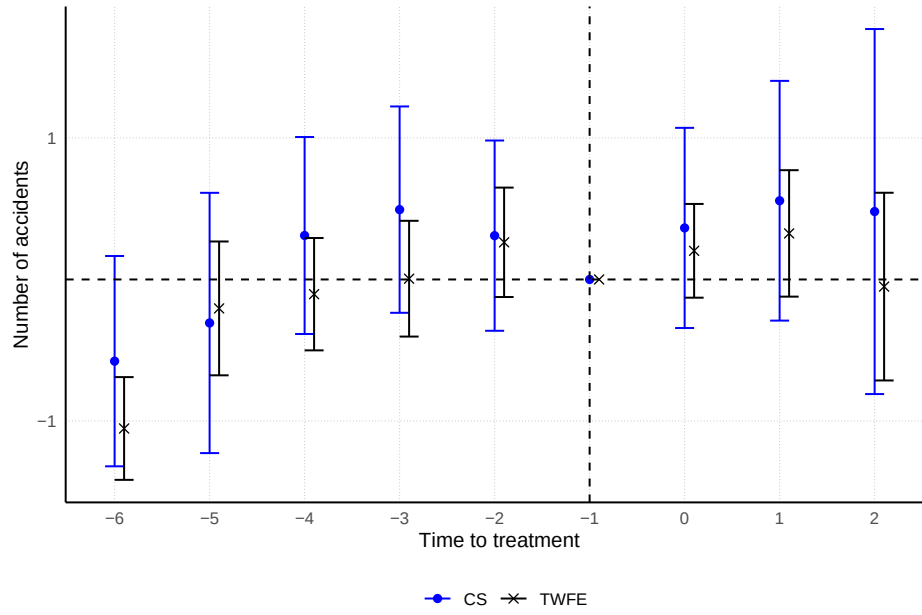
Notes: This figure shows the event-study aggregation of the estimated impacts of the program on school level accidents, relative to the year before the program. The data comes from the *Sistema de Alertas*. We compare earlier cohorts of entrants with not-yet-treated and never-treated schools. 95% confidence intervals reported.

Figure 11: Treatment Timing Map - NYT Baseline (Excluding 2020-2021)



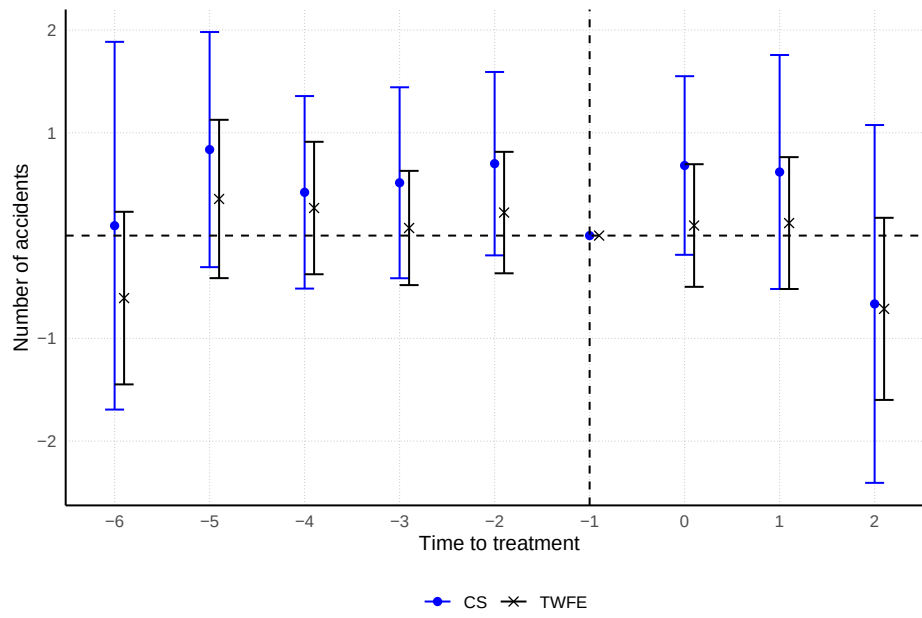
Notes: This figure shows the spatial distribution of treated and not-yet-treated (NYT) census tracts in Bogotá, by treatment cohort. Census tracts are classified as treated if their area intersects the catchment area of an active Ciempiés school path by at least 50%. The years 2020 and 2021 are excluded because the program was not fully operating as a walking-to-school caravan during the COVID-19 pandemic. The NYT cohort includes census tracts where the program had not yet started by 2023, so the census tracts where the program expanded in 2024 and 2025.

Figure 12: Event Study: Accidents Ages 13-18



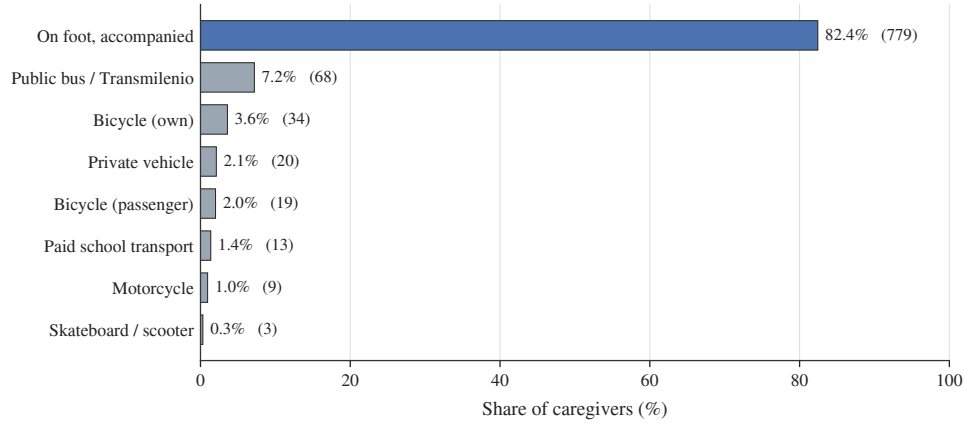
Notes: This figure shows event-study estimates of the impact of Ciempiés on traffic accidents involving adolescents aged 13 to 18, a group not directly targeted by the program. We show both the [Callaway and Sant'Anna \(2021\)](#) and the two-way fixed effects (TWFE) estimators. The control group consists of not-yet-treated census tracts. The years 2020 and 2021 are excluded because the program was not fully operating during the COVID-19 pandemic. 95% confidence intervals reported.

Figure 13: Event Study: Accidents Ages 5-12



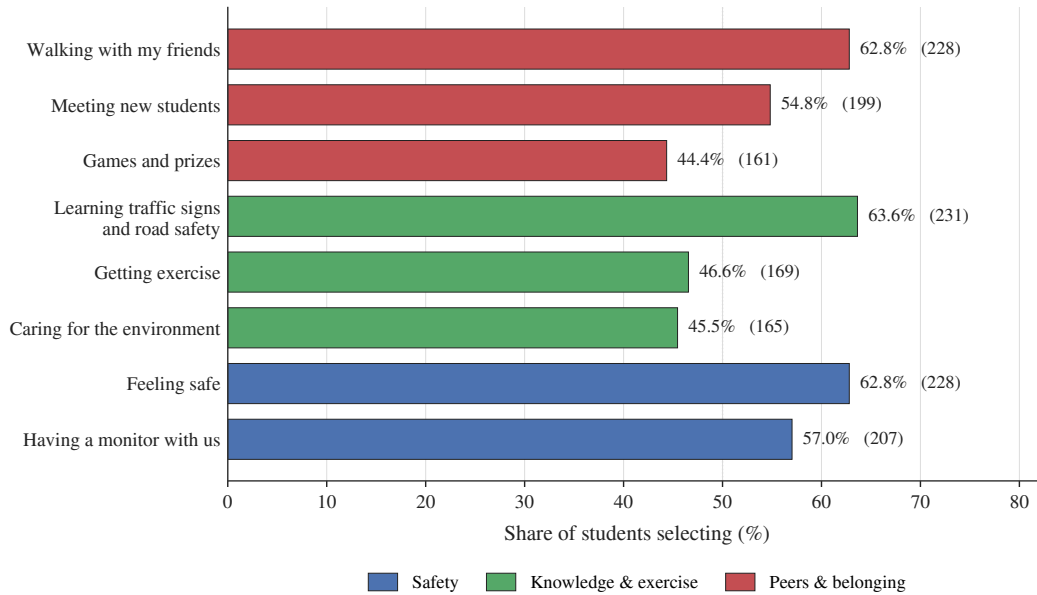
Notes: This figure shows event-study estimates of the impact of Ciempiés on traffic accidents involving children aged 5 to 12, the age group targeted by the program. We show both the [Callaway and Sant’Anna \(2021\)](#) and the two-way fixed effects (TWFE) estimators. The control group consists of not-yet-treated census tracts. The years 2020 and 2021 are excluded because the program was not fully operating during the COVID-19 pandemic. 95% confidence intervals reported.

Figure 14: Counterfactual School Commute Mode of *Ciempiés* Children



Notes: This figure shows how *Ciempiés* beneficiaries travelled to school *before* entering the program, as reported by their caregivers in the 2025 end-of-program survey ($n = 945$ caregivers). Bars show the share of caregivers selecting each mode and the underlying counts in parentheses. The vast majority report that the child walked, in nearly all cases accompanied by a parent or relative; motorized alternatives—public transport, paid school transport, private vehicles—account for less than 11% of the distribution combined.

Figure 15: Student-Reported Reasons for Liking *Ciempiés*, by Hypothesized Mechanism



Notes: This figure reports the share of student respondents in the 2025 end-of-program survey ($n = 363$) who selected each reason for enjoying the program (multi-select item). Colors map options onto the three student-facing mechanisms named in Section 1: safety; knowledge and exercise; peers and belonging. Counts shown in parentheses next to each bar.

Table 1: Descriptive Statistics for Ciempiés Children, 2021–2024

	2021	2022	2023	2024
Age	8.741 (1.996)	9.324 (2.421)	9.359 (2.505)	9.395 (2.273)
Grade	3.276 (1.839)	3.778 (2.272)	3.775 (2.336)	3.785 (2.089)
Share female	0.529 (0.499)	0.529 (0.499)	0.513 (0.500)	0.505 (0.500)
Socioeconomic stratum	2.069 (1.076)	1.982 (1.311)	2.051 (1.446)	2.113 (1.503)
Share victim conflict	0.035 (0.184)	0.042 (0.200)	0.090 (0.286)	0.077 (0.267)
Share Venezuelan	0.095 (0.293)	0.122 (0.327)	0.150 (0.358)	0.133 (0.339)
N	771	2139	2193	2820

Notes: This table shows means and standard deviations of student characteristics in our sample of Ciempies children, for the years 2021–2024.

Table 2: Descriptive Statistics for School Paths, 2020–2024

	2020	2021	2022	2023	2024
Length (km)	1.486 (0.392)	1.643 (0.481)	1.648 (0.354)	1.570 (0.361)	1.513 (0.305)
Walking time (min)	38.286 (9.736)	41.389 (8.558)	41.500 (8.767)	41.890 (7.176)	40.915 (6.452)
Share morning shift	0.571 (0.514)	0.481 (0.509)	0.475 (0.506)	0.439 (0.502)	0.434 (0.500)
Share afternoon shift	0.214 (0.426)	0.407 (0.501)	0.425 (0.501)	0.439 (0.502)	0.434 (0.500)
Share full shift	0.214 (0.426)	0.111 (0.320)	0.100 (0.304)	0.122 (0.331)	0.132 (0.342)
N	14	27	40	41	53

Notes: This table shows means and standard deviations of school path characteristics, for the years 2020–2024. Length is measured in kilometers and walking time in minutes. Shift shares represent the proportion of paths in each shift (morning, afternoon, or full-day).

Table 3: LPM: School Switching and Future Dropout

	Dropout (1)	Dropout (2)
School switching (L1)	0.028*** (0.001)	0.016*** (0.001)
School switching (L2)	0.032*** (0.001)	0.020*** (0.001)
Grade repeater (L1)	0.106*** (0.004)	0.067*** (0.003)
Grade repeater (L2)	0.049*** (0.004)	0.009*** (0.002)
Controls	No	Yes
FE: school $\times$ grade	Yes	Yes
Observations	3 094 788	1 520 258
Within R^2	0.009	0.031

Notes: The dependent variable (dropout) equals one if a student is enrolled in year t but does not re-enrol in year $t + 1$. The key regressors are indicators for school switching and grade repetition, entered at lag 1 (current year t) and lag 2 (prior year $t - 1$). All specifications include school $\times$ grade fixed effects; standard errors are clustered at the school level (in parentheses). Column (2) adds controls for sex, age, nationality, and SISBEN IV socioeconomic group (A–D); because SISBEN IV data are available only from 2022 onward, the sample with controls is smaller. *** $p < 0.01$.

Table 4: LPM: Branch Switching and Future Dropout

	Dropout (1)	Dropout (2)
Branch switching (L1)	0.021*** (0.001)	0.014*** (0.001)
Branch switching (L2)	0.021*** (0.001)	0.014*** (0.001)
Grade repeater (L1)	0.108*** (0.004)	0.068*** (0.003)
Grade repeater (L2)	0.050*** (0.005)	0.010*** (0.002)
Controls	No	Yes
FE: school $\times$ grade	Yes	Yes
Observations	3 094 788	1 520 258
Within R^2	0.008	0.031

Notes: The dependent variable (dropout) equals one if a student is enrolled in year t but does not re-enrol in year $t + 1$. Branch switching refers to moving between campuses (*sedes*) of the same school. The key regressors are indicators for branch switching and grade repetition, entered at lag 1 (current year t) and lag 2 (prior year $t - 1$). All specifications include school $\times$ grade fixed effects; standard errors are clustered at the school level (in parentheses). Column (2) adds controls for sex, age, nationality, and SISBEN IV socioeconomic group (A–D); because SISBEN IV data are available only from 2022 onward, the sample with controls is smaller. *** $p < 0.01$.

Table 5: Covariate Balance Statistics

variable	Unweighted			Weighted		
	Non-Adopt	Adopt	Norm. Diff.	Non-Adopt	Adopt	Norm. Diff.
Gender	0.482	0.490	0.103	0.503	0.495	-0.080
Stratum	1.933	2.007	0.270	1.907	1.984	0.285
Number of accidents	0.313	0.158	-0.259	0.575	0.202	-0.511
Insufficient (math)	0.453	0.394	-0.516	0.443	0.392	-0.488
Minimum (math)	0.312	0.344	0.483	0.314	0.337	0.405
Satisfactory (math)	0.151	0.172	0.453	0.154	0.176	0.494
Advanced (math)	0.084	0.090	0.120	0.089	0.095	0.125
Insufficient (language)	0.119	0.108	-0.189	0.107	0.101	-0.120
Minimum (language)	0.441	0.421	-0.288	0.431	0.420	-0.165
Satisfactory (language)	0.321	0.339	0.256	0.331	0.337	0.107
Advanced (language)	0.118	0.131	0.235	0.132	0.142	0.186

Notes: This table shows how the 43 school branches that adopted Ciempiés between 2019 and 2024 compare to the 75 branches that did not adopt it by 2024. We present the means and standardized differences for each variable by adoption status, both unweighted and weighted by the number of primary school students in each branch. All variables are measured as shares, except for stratum, which is measured on a scale of 1–6, and number of accidents. All variables are measured in 2018 (the year before the first cohort adopted Ciempiés), with the exception of Saber variables, which are measured in 2017. This was the last time that standardized test scores were applied to 5th grade children in Colombia.

Table 6: Dynamic ATTs by Outcome

	Repetition	Dropout	Switch School	Switch Sede
ATT(-5)	-0.043 (0.019)	0.000	-0.004 (0.038)	0.035 (0.048)
ATT(-4)	-0.014 (0.024)	0.000	-0.067 (0.031)	-0.028 (0.056)
ATT(-3)	-0.020 (0.025)	0.000	-0.051 (0.030)	-0.048 (0.047)
ATT(-2)	0.020 (0.025)	0.000	-0.023 (0.031)	0.032 (0.049)
ATT(0)	-0.035 (0.029)	-0.013 (0.003)	0.000 (0.035)	0.013 (0.053)
ATT(1)	-0.043 (0.032)	0.018 (0.023)	-0.048 (0.031)	-0.073 (0.047)
ATT(2)	-0.103 (0.024)	-0.046 (0.042)	-0.104 (0.021)	-0.159 (0.045)
ATT(3)	-0.018 (0.060)		-0.121 (0.094)	-0.219 (0.113)
N	31 639	41 192	31 639	31 639
Time Periods	9	8	8	8
Cohorts	4	3	4	4

Notes: Standard errors in parentheses. ATT estimates from dynamic difference-in-differences model using the Callaway and Sant’Anna (2021) method. Negative (positive) numbers indicate periods before (after) treatment. Control group consists of never-treated + not-yet-treated units.

Table 7: Counterfactual School Commute Mode of *Ciempién*s Children, 2025

	Caregivers (<i>n</i>)	Share %
On foot, accompanied by parent or relative	779	82.4
Public bus / Transmilenio / SITP	68	7.2
Bicycle (own)	34	3.6
Private vehicle	20	2.1
Bicycle (passenger of accompanying adult)	19	2.0
Paid school transport (<i>ruta escolar</i>)	13	1.4
Motorcycle	9	1.0
Skateboard / scooter / skates	3	0.3
Total	945	100.0

Notes: This table reports the school commute mode used by *Ciempién*s beneficiaries before entering the program, as reported by their caregivers. Sample: caregivers of currently enrolled children surveyed by the Secretary of Education of Bogotá, November–December 2025. Respondents selected one mode.

“Safe Paths to School: Evidence from Bogotá’s Walking to School Caravans”

Online Appendix

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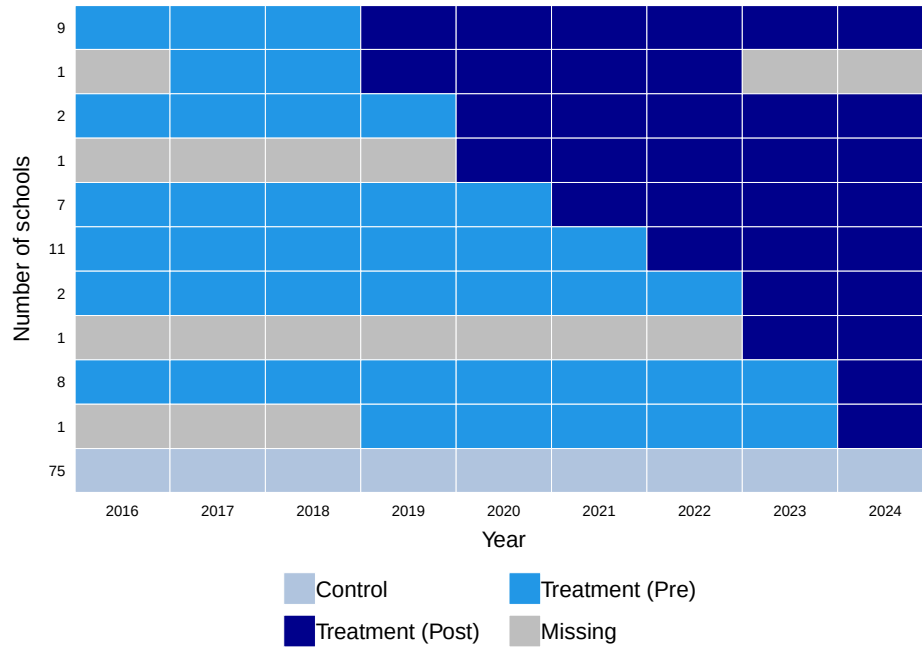
A Figures and Tables

Figure A.1: Example of Walking to School Caravan



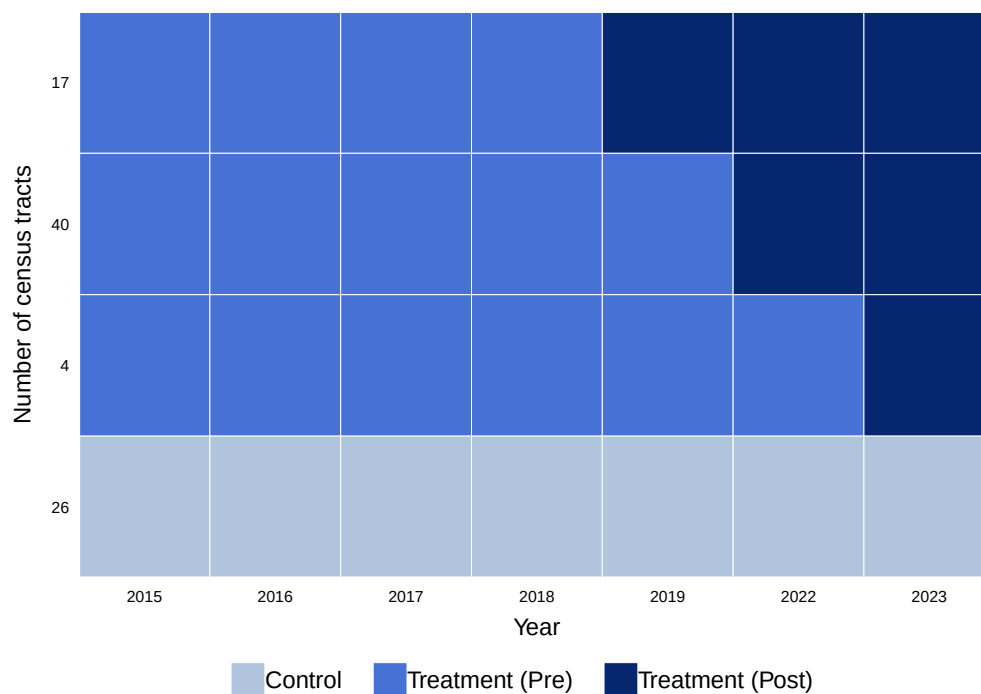
Notes: This figure shows photos from two Walking to School Caravans. The caravans are supervised by two adults, who ensure a safe commute to school. In the left panel we can see that these monitors often stop traffic for children to be able to cross the road.

Figure A.2: Treatment Rollout (Schools)



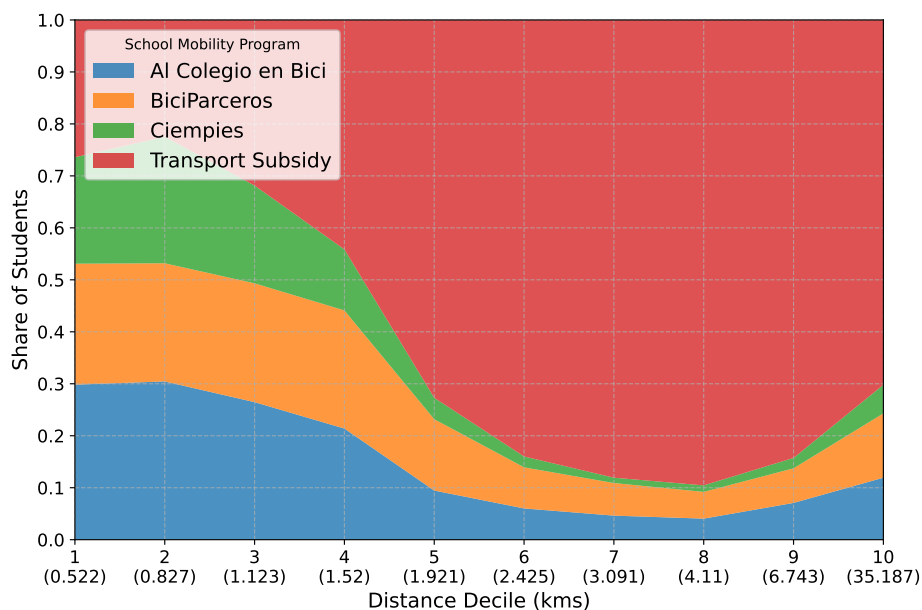
Notes: This figure shows the treatment rollout in schools from 2016 to 2024, marking the pre and post treatment periods for each cohort. The missing periods correspond to schools that opened or closed at some point during the sample period, and thus are not observed throughout the 9 years.

Figure A.3: Treatment Rollout (Census Tracts)



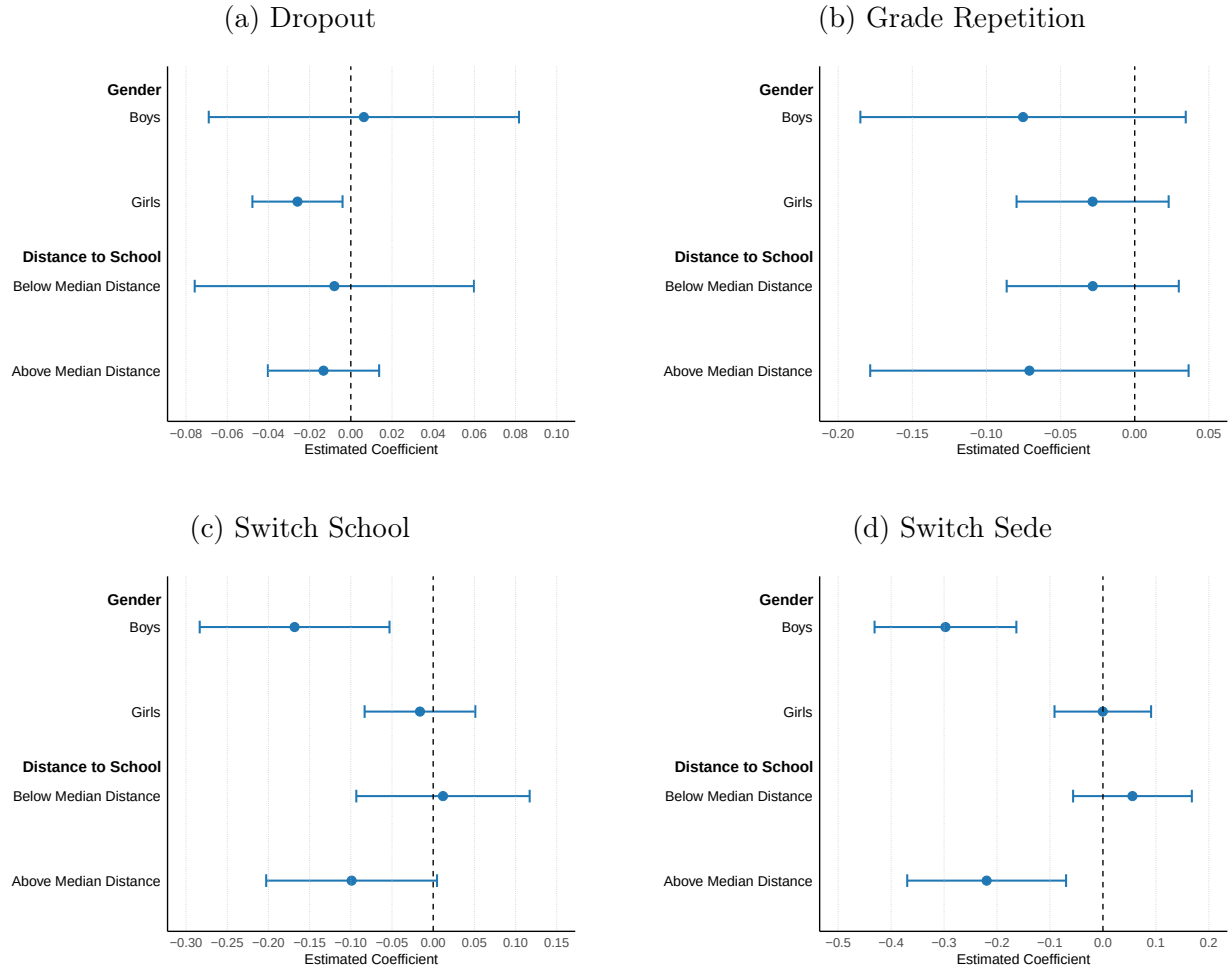
Notes: This figure shows the treatment rollout at the census tract level from 2015 to 2023, marking the pre and post treatment periods for each cohort. We exclude 2020 and 2021, since those years the program was not fully operating as walking-to-school caravan because of the covid pandemic.

Figure A.4: Shares of Mobility Programs by Distance to School in 2024



Notes: This figure shows the shares of students in the different mobility programs by decile of distance to school. The max value of the distance to school in each decile is reported in parenthesis.

Figure A.5: Effects of Walking to School Caravans by Children Attributes



Notes: This figure shows the overall ATT for the main four outcomes: dropout, grade repetition, school switching and sede switching, and different subsamples of children: boys, girls, living above median distance to school, and living below median distance to school. The control group is the same across all subsamples. 95% confidence intervals reported.

Table A.2: Description of Saber 5 Performance Levels

Level	Description
Advanced	Shows outstanding performance in the expected competencies for the evaluated subject area and grade.
Satisfactory	Demonstrates adequate performance in the required competencies for the subject area and grade. This is the level of performance that all or most students should reach.
Minimum	Shows minimum performance in the required competencies for the subject area and grade.
Insufficient	Does not surpass the most basic questions in the test.

Notes: This table shows the performance levels corresponding to the Saber 5 assessment scale. Saber 5 is a standardized test administered to fifth-grade students in Colombia to evaluate competencies in math and language.

Table A.1: Mechanism Evidence from End-of-Program Surveys, 2025

	Statistic	Value	Sample
<i>Panel A. Counterfactual mode and parent rotation at the meeting point</i>			
<i>A.1. Pre-Ciempíes school commute mode</i>			
On foot, accompanied by parent or relative	share %	82.4	Caregivers, $n=945$
Public bus / Transmilenio / SITP	share %	7.2	Caregivers
Bicycle (own)	share %	3.6	Caregivers
Private vehicle	share %	2.1	Caregivers
Paid school transport	share %	1.4	Caregivers
<i>A.2. Within-household exposure</i>			
Own children enrolled in Ciempíes	mean (sd)	1.28 (0.75)	Caregivers
Caregivers with ≥ 2 enrolled children	share %	21.6	Caregivers
<i>A.3. Parent rotation at the meeting point</i>			
Caregivers responsible for additional non-own children	share %	26.1	Caregivers
Additional non-own children supervised, conditional	mean (sd)	2.41 (2.60)	Caregivers, $n=207$
Additional non-own children supervised, conditional	median	2	Caregivers
<i>Panel B. Perceptions and attachment</i>			
<i>B.1. Subjective ratings (1–10 scale)</i>			
Happiness with the walk to school	mean (sd)	9.58 (1.18)	Students, $n=363$
Perceived safety with the monitors	mean (sd)	9.71 (0.80)	Students
Quality of monitor service	mean (sd)	9.81 (0.66)	Caregivers, $n=945$
Quality of caregiver–monitor interaction	mean (sd)	9.67 (1.00)	Caregivers
<i>B.2. Retention intent (next year)</i>			
Students wanting to continue	share %	97.8	Students
Caregivers wanting their child to continue	share %	98.2	Caregivers
<i>B.3. Behavioural changes since enrolment (caregivers)</i>			
“Goes to school with enthusiasm”	share %	17.4	Caregivers
“Is happier”	share %	14.3	Caregivers
“Is more responsible”	share %	12.3	Caregivers
“Learned traffic signals”	share %	11.4	Caregivers
“Knows the neighborhood and city better”	share %	11.1	Caregivers
<i>B.4. Most-cited reasons students like the program</i>			
Learning traffic signs and road safety	share %	63.6	Students
Walking with friends	share %	62.8	Students
Feeling safe	share %	62.8	Students
Having a monitor with us	share %	57.0	Students
Meeting new students	share %	54.8	Students

Notes: End-of-year surveys collected by the Secretary of Education of Bogotá in November–December 2025 from currently enrolled *Ciempíes* beneficiaries and their caregivers. **Because the instrument was administered to currently enrolled beneficiaries only, the sample is highly selected: switchers, dropouts, and non-participants are mechanically absent. All figures should therefore be interpreted with caution and cannot be read as causal estimates.** Caregivers: $n = 945$. Students: $n = 363$. Multi-select items in Panels B.3 and B.4 report the percentage of valid respondents who selected each option, so column entries can sum to more than 100%. Counts in Panel A are obtained from free-text responses parsed to integers; values above twenty are flagged as missing on the grounds that they are most likely caregiver identification numbers mistakenly entered in a count field. Panel A.3 reflects a conditional question: caregivers were first asked whether they were responsible at the meeting point for any children besides their own, and only those who answered yes were asked how many. The reported counts therefore refer to children *in addition to* the caregiver’s own.

B List of Acronyms

LMICs Low- and Middle-Income Countries	2
WSB Walking to School Buses	8
WSC Walking to School Caravans	2
DiD Difference-in-Differences	10
CCT Conditional Cash Transfers	4
ATT Average Treatment Effect on the Treated	10
QLS Quality of Life Survey	3